

IDW and ILDW (Short Pseudocode)

Notation. Each link ℓ is the segment $[p_{\ell,1}, p_{\ell,2}]$ with endpoints $p_{\ell,1}, p_{\ell,2} \in \mathbb{R}^2$, midpoint $m_\ell = (p_{\ell,1} + p_{\ell,2})/2$, and length $L_\ell = \|p_{\ell,2} - p_{\ell,1}\|_2$. Each pixel p has center $c_p \in \mathbb{R}^2$.

IDW

Algorithm 1 Discretized Midpoint IDW

Require: links $(p_{\ell,1}, p_{\ell,2}, A_\ell, f_\ell, \text{pol}_\ell)$, radius $r_{\max}$

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1: for each link  $\ell$  do
2:    $L_\ell \leftarrow \|p_{\ell,2} - p_{\ell,1}\|_2$ 
3:    $(\kappa_\ell, \alpha_\ell) \leftarrow \text{ITU838}(f_\ell, \text{pol}_\ell)$ 
4:    $R_\ell \leftarrow \left( \frac{A_\ell/L_\ell}{\kappa_\ell} \right)^{1/\alpha_\ell}$ 
5: end for
6: for each pixel  $p$  with center  $c_p$  do
7:    $m_\ell \leftarrow (p_{\ell,1} + p_{\ell,2})/2$  for each link  $\ell$ 
8:   if one or more link midpoints lie inside pixel  $p$  then
9:      $\hat{R}_p \leftarrow$  mean of  $R_\ell$  of those links
10:  else
11:     $N_p \leftarrow \{\ell : \|m_\ell - c_p\|_2 \leq r_{\max}\}$ 
12:    if  $N_p = \emptyset$  then
13:       $\hat{R}_p \leftarrow 0$ 
14:    else
15:       $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \frac{\|m_\ell - c_p\|_2^{-2}}{\sum_{j \in N_p} \|m_j - c_p\|_2^{-2}} R_\ell$ 
16:    end if
17:  end if
18: end for
19: Return field  $\hat{R}$ 

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ILDW

Algorithm 2 Segment-Distance ILDW

Require: links $(p_{\ell,1}, p_{\ell,2}, A_{\ell}, f_{\ell}, \text{pol}_{\ell})$, radius $r_{\max}$

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1: for each link  $\ell$  do
2:    $L_{\ell} \leftarrow \|p_{\ell,2} - p_{\ell,1}\|_2$ 
3:    $(\kappa_{\ell}, \alpha_{\ell}) \leftarrow \text{ITU838}(f_{\ell}, \text{pol}_{\ell})$ 
4:    $R_{\ell} \leftarrow \left( \frac{A_{\ell}/L_{\ell}}{\kappa_{\ell}} \right)^{1/\alpha_{\ell}}$ 
5: end for
6: for each pixel  $p$  with center  $c_p$  do
7:   if one or more links cross pixel  $p$  then
8:      $\hat{R}_p \leftarrow$  mean of  $R_{\ell}$  of those links
9:   else
10:     $N_p \leftarrow \{\ell : \text{dist}(c_p, [p_{\ell,1}, p_{\ell,2}]) \leq r_{\max}\}$ 
11:    if  $N_p = \emptyset$  then
12:       $\hat{R}_p \leftarrow 0$ 
13:    else
14:       $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \frac{\text{dist}(c_p, [p_{\ell,1}, p_{\ell,2}])^{-2}}{\sum_{j \in N_p} \text{dist}(c_p, [p_{j,1}, p_{j,2}])^{-2}} R_{\ell}$ 
15:    end if
16:  end if
17: end for
18: Return field  $\hat{R}$ 

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